

Practical # 2: Maven + Docker Integration

Step-01: Create a folder and get into that folder

```
VICTUS@Jarvis MINGW64 ~ (master)
$ cd docker-maven-prac

VICTUS@Jarvis MINGW64 ~/docker-maven-prac (master)
$ mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClass
Definer (file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom -----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO]
[INFO] Using following parameters for creating project from old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] -----
[INFO] Parameter: basedir, Value: C:\Users\VICTUS\docker-maven-prac
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: docker-demo
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from old (1.x) Archetype in dir: C:\Users\VICTUS\docker-maven-prac\docker-demo
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 1.235 s
[INFO] Finished at: 2026-05-03T16:57:54+05:30
[INFO]
[INFO] -----
```

STEP-02: Create maven project

mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

STEP-03: Get inside the project and check the directory

STEP-04: Make changes in the pom.xml, create a dockerfile and then rename it.

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    < http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>

<!-- Must match your archetype command -->

<groupId>com.example</groupId>

<artifactId>docker-demo</artifactId>

<version>1.0-SNAPSHOT</version>

<packaging>jar</packaging>

<name>docker-demo</name>

<!-- Java Version -->

<properties>

    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>

    <maven.compiler.source>21</maven.compiler.source>

    <maven.compiler.target>21</maven.compiler.target>
```

```
</properties>

<!-- Dependencies -->
<dependencies>

  <!-- JUnit -->
  <dependency>

    <groupId>junit</groupId>

    <artifactId>junit</artifactId>

    <version>4.13.2</version>

    <scope>test</scope>

  </dependency>
</dependencies>

<!-- Build Plugins -->
<build>

  <plugins>

    <!-- Executable JAR -->
    <plugin>

      <groupId>org.apache.maven.plugins</groupId>

      <artifactId>maven-jar-plugin</artifactId>

      <version>3.3.0</version>

      <configuration>

        <archive>

          <manifest>

            <mainClass>com.example.App</mainClass>

          </manifest>

        </archive>

      </configuration>

    </plugin>

    <!-- Docker Plugin -->
    <plugin>

      <groupId>com.spotify</groupId>

      <artifactId>dockerfile-maven-plugin</artifactId>

      <version>1.4.13</version>

      <configuration>

        <!-- Must match project name -->

        <repository>docker-demo</repository>

      </configuration>

    </plugin>

  </plugins>

</build>

</project>
```

```

        <tag>${project.version}</tag>

        <!-- Important -->

        <dockerfile>${project.basedir}/Dockerfile</dockerfile>

    </configuration>

    <executions>

        <execution>

            <id>default</id>

            <goals>

                <goal>build</goal>

            </goals>

        </execution>

    </executions>

</plugin>

</plugins>

</build>

</project>

```

Dockerfile

FROM eclipse-temurin:21-jdk

WORKDIR /app

COPY target/docker-demo-1.0-SNAPSHOT.jar app.jar

ENTRYPOINT [""java","-jar","app.jar","]

```

VICTUS@jarvis MINGW64 ~/docker-maven-prac (master)
$ cd docker-demo

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo (master)
$ dir
pom.xml  src

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo (master)
$ notepad pom.xml

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo (master)
$ notepad dockerfile

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo (master)
$ cd src

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src (master)
$ cd main

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src/main (master)
$ cd java

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src/main/java (master)
$ cd com

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src/main/java/com (master)
$ cd example

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src/main/java/com/example (master)
$ notepad App.java

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src/main/java/com/example (master)
$ cd ..

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src/main/java/com (master)
$ cd ..

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src/main/java (master)
$ cd ..

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src/main (master)
$ cd ..

VICTUS@jarvis MINGW64 ~/docker-maven-prac/docker-demo/src (master)
$ cd ..

```

STEP-05: Modify Java Program with path : src/main/java/com/example/App.java **STEP-06:**
Build the package

```
Command Prompt
[INFO] ---> d552edd869aa
[INFO] Successfully built d552edd869aa
[INFO] Successfully tagged docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] Detected build of image with id d552edd869aa
[INFO] Building jar: C:\Users\narut\docker-maven2\docker-demo\target\docker-demo-1.0-SNAPSHOT-docker-info.jar
[INFO] Successfully built docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 31.998 s
[INFO] Finished at: 2026-03-27T01:04:10+05:30
[INFO]
C:\Users\narut\docker-maven2\docker-demo>docker run docker-demo:1.0-SNAPSHOT
Hello from Docker Maven Demo!

C:\Users\narut\docker-maven2\docker-demo>docker images
```

U	In Use	IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
		alpine:latest	25109184c71b	13.1MB	3.95MB	U
		ashika0804/ab2:v1.0	28e750845649	476MB	148MB	
		ashika0804/ap1:v1	5c1c79cf906a	33.6MB	10.1MB	
		ashika0804/c1:v1.1	cd1dba651b30	119MB	31.7MB	U
		docker-demo:1.0-SNAPSHOT	d552edd869aa	675MB	211MB	U
		docker/welcome-to-docker:latest	c4d56c24da4f	22.2MB	6.03MB	U
		eclipse-temurin:21-jdk	3b0a98dfbdf1	679MB	214MB	
		ghcr.io/ashika07734/ghcr-demo:latest	25d6eacf3081	92.5MB	26MB	
		ghcr.io/ashika07734/sql_7739:v1	45abbc49f01a	784MB	166MB	U
		httpd:latest	dd178595edd6	177MB	47.6MB	U
		mongo:6	03cda579c8ca	1.06GB	273MB	U
		mysql:5.7	4bc6bc963e6d	700MB	149MB	
		mysql:8.0	aa956cd6c5fa	784MB	166MB	U
		mysql:8.1	a1361596e618	784MB	166MB	U
		mysql:latest	6b18d01fb632	1.27GB	283MB	U
		nginx:latest	c881927c4077	240MB	65.7MB	U
		node-mongo-compose-app:latest	a6292355b97b	1.63GB	406MB	U

STEP-07: Run the container **STEP-08:**

Verify the image